Prelims only Textual
Marks 80
Time :− 3hrs

CBSE : X
Date :

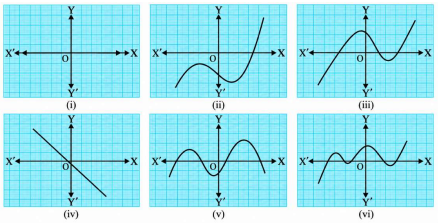
MATHS BRIDGE ACADEMY

Lodha Signet−A,425 4th Floor, Kolshet Rd , Thane − **9892005310**

SECTION A — 1 mark each (9 × 1 = 9)

1. Given that HCF(306, 657) = 9, find LCM(306, 657).

2. The graphs of $y = p(x)$ are given below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$ in each case.



3. Which term of the AP: 3, 8, 13, 18, ... is 78?

4. Find the distance between the points (2, 3) and (4, 1).

5. Fill in the blanks using the correct word given in brackets:

(i) All circles are ____. (congruent, similar)

(ii) All squares are ____. (similar, congruent)

6. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that OQ = 12 cm. Length PQ is:

(A) 12 cm (B) 13 cm (C) 8.5 cm (D) $\sqrt{119}$ cm

7. Choose the correct option and justify your choice: $2\tan 30° / (1 + \tan^2 30°) =$

(A) sin 60° (B) cos 60° (C) tan 60° (D) sin 30°

8. Tick the correct answer: Area of a sector of angle p (in degrees) of a circle with radius R is

(A) $\left(\frac{p}{180}\right) \times 2\pi R$ (B) $\left(\frac{p}{180}\right) \times \pi R^2$ (C) $\left(\frac{p}{360}\right) \times 2\pi R$ (D) $\left(\frac{p}{720}\right) \times 2\pi R^2$

9. Which of the following cannot be the probability of an event?

(A) 2/3 (B) −1.5 (C) 15% (D) 0.7

SECTION B — 2 marks each (15 × 2 = 30)

10. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers.

11. Find the zeroes of the quadratic polynomial $x^2 - 2x - 8$, and verify the relationship between the zeroes and the coefficients.

12. Find a quadratic polynomial, the sum and product of whose zeroes are 1/4 and −1 respectively.

13. On comparing the ratios $\frac{a^1}{a^2}, \frac{b^1}{b^2}, \frac{c^1}{c^2}$, find out whether the following pairs of linear equations are consistent or inconsistent:

(i) $3x + 2y = 5;\ 2x - 3y = 7$ (ii) $2x - 3y = 8;\ 4x - 6y = 9$

14. Find the nature of the roots of the quadratic equation $2x^2 - 3x + 5 = 0$. If the real roots exist, find them.

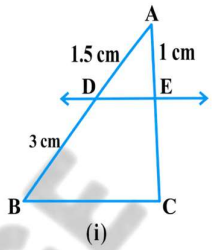
15. If the 3rd and the 9th terms of an AP are 4 and −8 respectively, which term of this AP is zero?

16. The 17th term of an AP exceeds its 10th term by 7. Find the common difference.

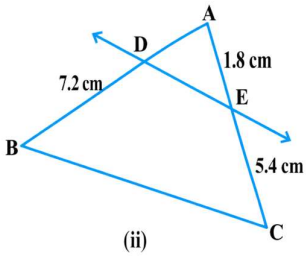
MATHS BRIDGE ACADEMY

17. Find the coordinates of the point which divides the line segment joining $(-1, 7)$ and $(4, -3)$ in the ratio 2 : 3.

18. In the figure, (i) and (ii), DE ∥ BC. Find EC in (i) and AD in (ii).



(i)



(ii)

19. The length of a tangent from a point A at distance 5 cm from the centre of a circle is 4 cm. Find the radius of the circle.

20. If sin A = ¾, calculate cos A and tan A.

21. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30°. Find the height of the tower.

22. A solid is in the shape of a cone standing on a hemisphere, with both their radii being equal to 1 cm, and the height of the cone equal to its radius. Find the volume of the solid in terms of π.

23. A die is thrown once. Find the probability of getting:
(i) a prime number (ii) a number lying between 2 and 6 (iii) an odd number

24. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is (i) red? (ii) not red?

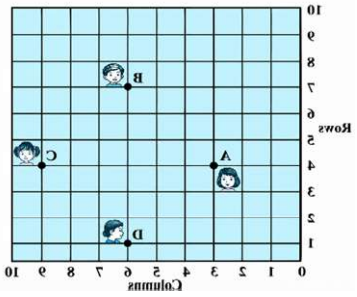
SECTION C — 3 marks each (11 × 3 = 33)

25. Prove that $\sqrt{5}$ is irrational.

26. Form the pair of linear equations for the following problem and find its solution by the substitution method: The coach of a cricket team buys 7 bats and 6 balls for ₹3800. Later, she buys 3 bats and 5 balls for ₹1750. Find the cost of each bat and each ball.

27. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

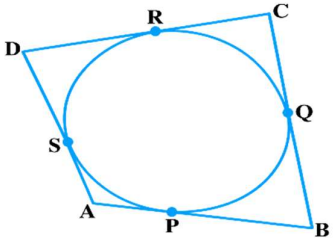
28. In a classroom, 4 friends are seated at the points A, B, C and D as shown in the figure. Champa and Chameli walk into the class and after observing for a few minutes, Champa asks Chameli, "Don't you think ABCD is a square?" Chameli disagrees. Using the distance formula, find who is correct.



29. ABCD is a trapezium in which AB ∥ DC and its diagonals intersect each other at the point O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$.

30. A vertical pole of length 6 m casts a shadow 4 m long on the ground, and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

31. A quadrilateral ABCD is drawn to circumscribe a circle (see figure). Prove that AB + CD = AD + BC.



32. Prove the identity: $(cosec\ \theta - \cot\theta)^2 = \frac{(1-\cos\theta)}{(1+\cos\theta)}$, where the angles involved are acute angles for which the expressions are defined.

33. A gulab jamun contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 gulab jamuns, each shaped like a cylinder with two hemispherical ends, with length 5 cm and diameter 2.8 cm.

34. The following distribution shows the number of plants in 20 houses of a locality. Find the mean number of plants per house. Which method did you use, and why?

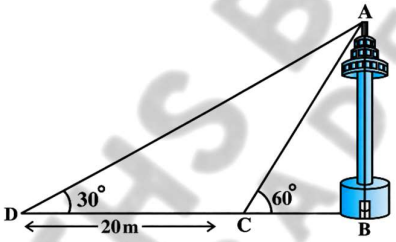
Number of plants	0–2	2–4	4–6	6–8	8–10	10–12	12–14
Number of houses	1	2	1	5	6	2	3

35. The following data gives the observed lifetimes (in hours) of 225 electrical components. Determine the modal lifetime of the components.

Lifetimes (hrs)	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	10	35	52	61	38	29

SECTION D — 4 marks each, applied / multi-part (2 × 4 = 8)

36. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from this point, on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30° (see figure). Find the height of the tower and the width of the canal.



37. A horse is tied to a peg at one corner of a square-shaped grass field of side 15 m, by means of a 5 m long rope (see figure). Find:

(i) the area of the field the horse can graze.

(ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$)

